

Information Technology University, Lahore

Applied Machine Learning BSCE23 Spring-2026 Assignment # 02

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1. Data Preparation Logic

The foundation of any medical diagnostic system is the quality and preparation of data.

- **Feature Engineering:** The dataset contains 30 numerical features (mean, standard error, and "worst" values for 10 cell nucleus characteristics). We mapped the diagnosis labels to a binary format where 0 represents Malignant and 1 represents Benign.
- **Data Splitting:** Using a 70:30 split with your Roll No (42) as the random seed ensures reproducibility. This split provides enough data for the models to learn patterns while keeping a significant portion (171 samples) aside to test how well the models generalize to unseen "patients."
- **Standardization:** This is the most critical preprocessing step. SVM calculates the distance between data points; if one feature (like Area) has values in the 1000s and another (like Smoothness) is 0.1, the model will ignore the smaller feature. Standardization brings all features to a common scale (Mean=0, Variance=1).

2. Model Implementation Breakdown

Naive Bayes (Gaussian)

- **How it works:** It uses the Bayes' Theorem to calculate the probability that a set of cell measurements belongs to a specific class.
- **The "Gaussian" Assumption:** Since our features are continuous (decimals), we assume the data follows a "Normal" or Bell Curve distribution.
- **The "Scratch" Logic:** In the manual code, we calculate the mean (μ) and variance (σ^2) for every feature for both malignant and benign cases. To predict, we calculate the likelihood of a new sample appearing under those curves and pick the class with the highest probability.

Support Vector Machines (SVM)

- The Objective: SVM looks for the Optimal Hyperplane—a boundary that leaves the maximum possible "margin" (gap) between the two classes.
- Linear Kernel: This assumes the two types of tumors can be separated by a straight line in high-dimensional space.
- RBF (Radial Basis Function) Kernel: If the data were "circular" or nested (non-linear), RBF projects the data into an even higher dimension to find a separation.
- The "Scratch" Logic: Our manual implementation used Hinge Loss and Gradient Descent. The model iteratively adjusts its weights to "penalize" points that fall on the wrong side of the boundary or too close to the margin.

3. Evaluation Metrics Explained

In medical systems, "Accuracy" alone isn't enough. We look at:

- Confusion Matrix: A table showing exactly where the model got confused. It tracks False Negatives (the most dangerous error in medicine—calling a malignant tumor "benign").
- Precision: Of all predicted "Benign" cases, how many were actually benign?
- Recall (Sensitivity): Of all the people who actually had "Malignant" tumors, how many did the model correctly identify?
- F1-Score: The harmonic mean of Precision and Recall, providing a single score for model balance.

4. Comparative Analysis (Report Summary)

Performance Comparison Table

METRIC	Naive Bayes	SVM (Linear/RBF)
Accuracy	~93.5%	~97.6%
Handling Correlated Features	Poor (assumes independence)	Excellent
Complexity	Low / Fast	High / Robust

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Recommendations

Based on the results (where SVM achieved 97.66% accuracy compared to 93.57% for Naive Bayes), the Support Vector Machine (SVM) is the clear recommendation for a clinical environment.

While Naive Bayes is faster, it is hampered by its "independence assumption"—in biology, things like cell radius, area, and perimeter are mathematically related. SVM ignores these correlations and focuses only on the "Support Vectors" (the most difficult-to-classify samples), leading to a much more reliable diagnostic tool.

OUTPUT:

	Model	Accuracy	Precision	Recall
0	NB Scratch	0.935673	0.935523	0.935673
1	NB Sklearn	0.935673	0.935523	0.935673
2	SVM Scratch	0.982456	0.982469	0.982456
3	SVM Linear	0.976608	0.976608	0.976608
4	SVM RBF	0.976608	0.976608	0.976608

